



LifeLink API

1.0.0 OAS 3.0

Blood donation matching platform — REST API documentation

Servers

https://graduation-project-cy61.onrender.com/api/v1 - Production (Render)

Authorize



Discovery

Public hospital discovery APIs



GET	/hospitals	List hospitals for discovery		
GET	/hospitals/nearby	List nearby hospitals		
GET	/hospitals/{id}	Get hospital details by id		

Help

Help and document endpoints



GET	/help/faq	Get FAQ list		
GET	/help/documents/{type}	Get help document by type		

Notifications

Authenticated user notifications



GET	/notifications	List notifications for authenticated user		
PATCH	/notifications/{id}/read	Mark one notification as read		
PATCH	/notifications/read-all	Mark all notifications as read		
GET	/notifications/{id}	Get one notification		
DELETE	/notifications/{id}	Delete one notification		

Rewards - Donor

Donor-facing reward endpoints



GET	/rewards/points	Get donor's points summary and tier		
GET	/rewards/points/history	Get donor's points transaction history		
GET	/rewards/badges	Get all badges and donor's progress		
GET	/rewards/catalog	Get available rewards catalog		
GET	/rewards/history	Get donor reward history		

POST	/rewards/catalog/{rewardId}/redeem	Redeem a reward using points		
GET	/rewards/redemptions	Get donor's redemption history		
GET	/rewards/leaderboard	Get the top donors leaderboard		

Rewards - Admin

Admin reward management

POST	/rewards/admin/users/{userId}/points/adjust	Manually adjust a donor's points (admin only)		
PATCH	/rewards/admin/catalog/{rewardId}/status	Update a reward's status (admin only)		
GET	/rewards/admin/analytics	Get rewards system analytics (admin only)		

Support

Support contact endpoint

POST	/support/contact	Submit a support request		
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Admin

GET	/admin/profile	Get admin profile		
GET	/admin/alerts	Get the admin alert summary		
GET	/admin/blood-inventory-summary	Get the system blood inventory summary		
GET	/admin/donors	List donors		
GET	/admin/hospitals	List hospitals		
GET	/admin/donors/{id}	Get a donor by id		
PUT	/admin/donors/{id}	Update a donor		
GET	/admin/hospitals/{id}	Get a hospital by id		
GET	/admin/admins	List admin users		
POST	/admin/admins	Create a new admin account		
GET	/admin/admins/{id}	Get an admin by id		
PUT	/admin/admins/{id}	Update an admin account		
DELETE	/admin/admins/{id}	Delete an admin account		
POST	/admin/donors/{id}/ban	Ban a donor		
POST	/admin/donors/{id}/unban	Unban a donor		
PUT	/admin/hospitals/{id}/status	Update hospital suspension status		

GET	/admin/permissions/roles	List role permissions		
POST	/admin/permissions/roles	Create a role permission record		
GET	/admin/permissions/roles/{role}	Get a role permission record		
PUT	/admin/permissions/roles/{role}	Update role permissions		

Admin - System

GET	/admin/system/health	System health check		
POST	/admin/system/maintenance	Toggle maintenance mode		
GET	/admin/system/maintenance	Get maintenance mode status		

Admin - Analytics

GET	/admin/statistics	Get admin statistics summary		
GET	/admin/analytics/dashboard	Dashboard summary with key metrics		
GET	/admin/analytics/donations	Monthly donation trends		
GET	/admin/analytics/blood-types	Blood type distribution		
GET	/admin/analytics/top-donors	Top donors leaderboard		
GET	/admin/analytics/growth	Growth metrics over time		

Admin - Audit

GET	/admin/audit-logs	List audit logs		
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Admin - Users



GET	/admin/users	List all users with filters and pagination		
GET	/admin/users/stats	Get user statistics		
POST	/admin/users/hospital	Create a new hospital (admin-created, pre-verified)		
GET	/admin/users/{id}	Get user details by ID		
DELETE	/admin/users/{id}	Soft-delete a user account		
PATCH	/admin/users/{id}/verify	Verify a user's email		
PATCH	/admin/users/{id}/unverify	Unverify a user's email		
PATCH	/admin/users/{id}/suspend	Suspend a user account		
PATCH	/admin/users/{id}/unsuspend	Unsuspend a user account		

Admin - Requests



GET	/admin/requests	List all blood/organ requests		
GET	/admin/requests/stats	Get request statistics		
GET	/admin/requests/{id}	Get full request details with donations		
GET	/admin/requests/{id}/donations	Get donations for a specific request		
PATCH	/admin/requests/{id}/fulfill	Mark a request as fulfilled		
PATCH	/admin/requests/{id}/cancel	Cancel a request		
POST	/admin/requests/{id}/broadcast	Broadcast request to eligible donors		

Admin - Emergency



POST	/admin/emergency/broadcast	Send emergency broadcast to donors		
GET	/admin/emergency/critical	List critical and high urgency active requests		
GET	/admin/emergency/shortage-alerts	Blood shortage alerts		

Appointments



POST	/donations/book-appointment	Book a donor appointment with a hospital		
GET	/donations/book-appointment/my-appointments	Get the authenticated donor appointments		
DELETE	/donations/book-appointment/{appointmentId}	Cancel a donor appointment		
GET	/donations/my-appointments	Get donor appointments via the donations compatibility alias		

Auth



POST	/auth/signup	Register a new donor or hospital account		
POST	/auth/login	Login with email and password		
POST	/auth/register	Register a new user (compatibility alias)		
POST	/auth/validate-token	Validate current JWT and return session state		
POST	/auth/send-otp	Send a password reset OTP		
POST	/auth/verify-otp	Verify password reset OTP		
POST	/auth/2fa/setup	Initialize 2FA setup		
POST	/auth/2fa/confirm-setup	Confirm 2FA setup for the authenticated user		
POST	/auth/2fa/verify	Complete login by verifying a 2FA code		
POST	/auth/2fa/disable	Disable 2FA		
POST	/auth/logout	Logout and blacklist refresh token		
POST	/auth/refresh-token	Issue a new access token from refresh token		
POST	/auth/forgot-password	Request password reset		
POST	/auth/reset-password	Reset password with token		
POST	/auth/verify-email	Send email verification message		
POST	/auth/verify-email-token	Verify email with token		
GET	/auth/me	Get current authenticated user		
POST	/auth/fcm-token	Register the current device FCM token		
PUT	/auth/fcm-token	Replace the stored FCM tokens with the current active device token		
DELETE	/auth/fcm-token	Remove a device FCM token		

Donations



POST	/donations/complete	Mark a donation as completed		
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Donor

GET	/donor/profile	Get the authenticated donor profile			
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PUT	/donor/profile	Update the authenticated donor profile		
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GET	/donor/requests	List active donation requests available to donors		
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GET	/donor/matches	List donation requests matched to the authenticated donor		
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POST	/donor/respond/{requestId}	Respond to a donation request		
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GET	/donor/history	List donation history for the authenticated donor		
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PUT	/donor/availability	Update donor availability		
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GET	/donor/health-history	Get the authenticated donor health history		
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PATCH	/donor/health-history	Update the authenticated donor health history		
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Hospital

GET	/hospital/profile	Get the authenticated hospital profile		
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PUT	/hospital/profile	Update the authenticated hospital profile		
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POST	/hospital/request	Create a donation request		
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GET	/hospital/requests	List requests created by the authenticated hospital		
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GET	/hospital/requests/{requestId}	Get details for a specific hospital request		
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PUT	/hospital/requests/{requestId}	Update the status of a hospital request		
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DELETE	/hospital/requests/{requestId}	Cancel a hospital request		
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GET	/hospital/donations	List donations for the authenticated hospital's requests		
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GET	/hospital/blood-inventory	Get a read-only blood inventory summary for the authenticated hospital		
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GET	/hospital/blood-bank-settings	Get hospital blood bank settings		
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PUT	/hospital/blood-bank-settings	Update hospital blood bank settings		
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GET	/hospital/notification-preferences	Get hospital notification preferences		
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PUT	/hospital/notification-preferences	Update hospital notification preferences		
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GET	/hospital/reports/monthly	Get monthly hospital report		
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GET	/hospital/staff	List hospital staff		
POST	/hospital/staff	Create hospital staff member		
DELETE	/hospital/staff/{id}	Delete hospital staff member		

Schemas	
BaseUser	>
SignupDonorRequest	>
SignupHospitalRequest	>
LoginRequest	>
RefreshTokenRequest	>
EmailRequest	>
TokenRequest	>
OtpRequest	>
AuthTokens	>
FcmTokenRequest	>
ErrorResponse	>